COMPLEX OF WASTE DISPOSAL DEVICES. SMART BIN

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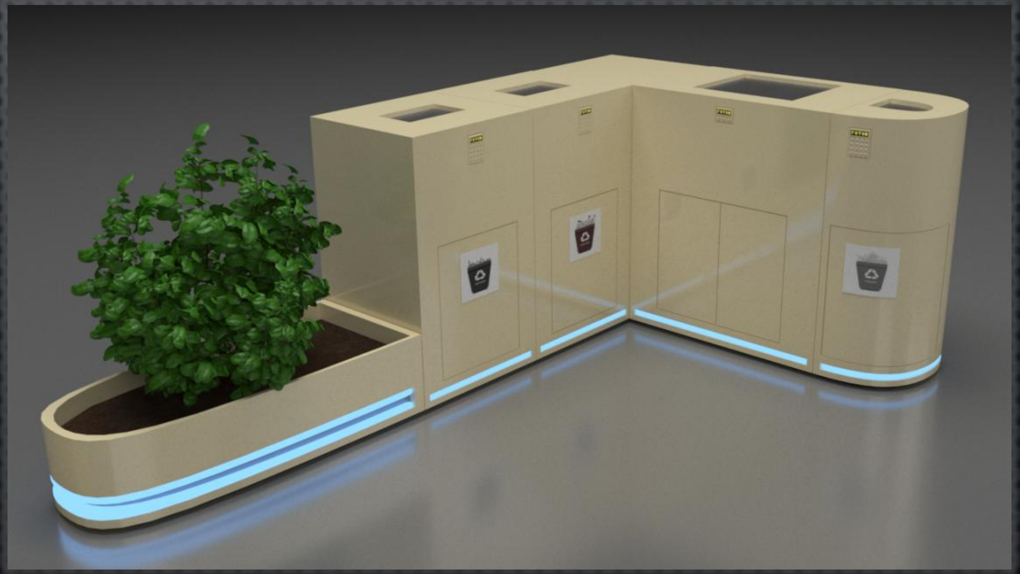
We present the design of a compact waste recycling machine. It is designed for everyday use by citizens to sort household waste (glass, batteries, and other waste) and shred it (metal and plastic).

The device is convenient because it takes up little space and can be placed either outdoors (preferably under a roof in a specially prepared area) or indoors. It can be docked to one or more similar devices, creating a multifunctional unit that allows for the collection and disposal of various types of waste in a single location.



The device has an attractive, minimalist design. The body is made of impact-resistant glossy plastic and features a neon light underneath. The height is 960 mm, the depth is 500 mm, and the width can be 320 mm, 550 mm, or 800 mm, depending on the type of waste disposal unit. The electronic control panel is located on the front of the unit, at the top, and features input/control buttons and a monochrome display. The top of the unit features an opening covered by a flap where waste is placed. A door at the bottom of the front of the unit allows access to the interior, allowing removal of the waste container and access to the electronics and mechanisms.

An additional design element is the addition of a miniature plant bed (with backlighting, like the other elements). Besides serving as a decorative element, it symbolizes ecology – one of the reasons for collecting and recycling waste.

This device can be divided into several types.

The first type includes a mechanical component, tracks the amount of recycled material, and is used for:

- recycling and shredding (or crushing) metal

- recycling and shredding plastic

The second type eliminates the mechanical component that shreds waste, tracks the amount of recycled material, and is used for:

- battery recycling

- glass recycling

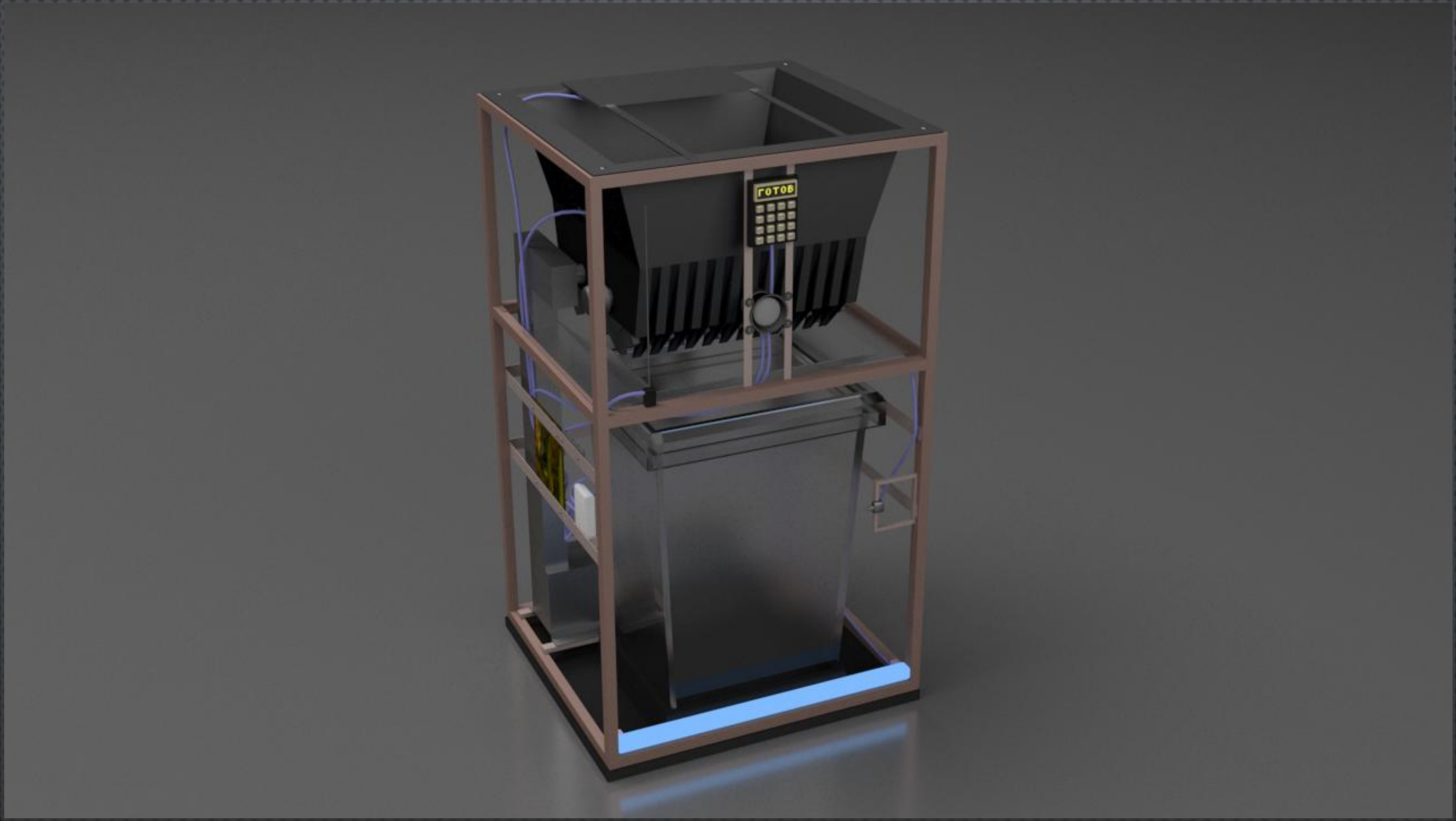
The third type has minimal electronics and functions as a standard trash can.

METAL, PLASTIC

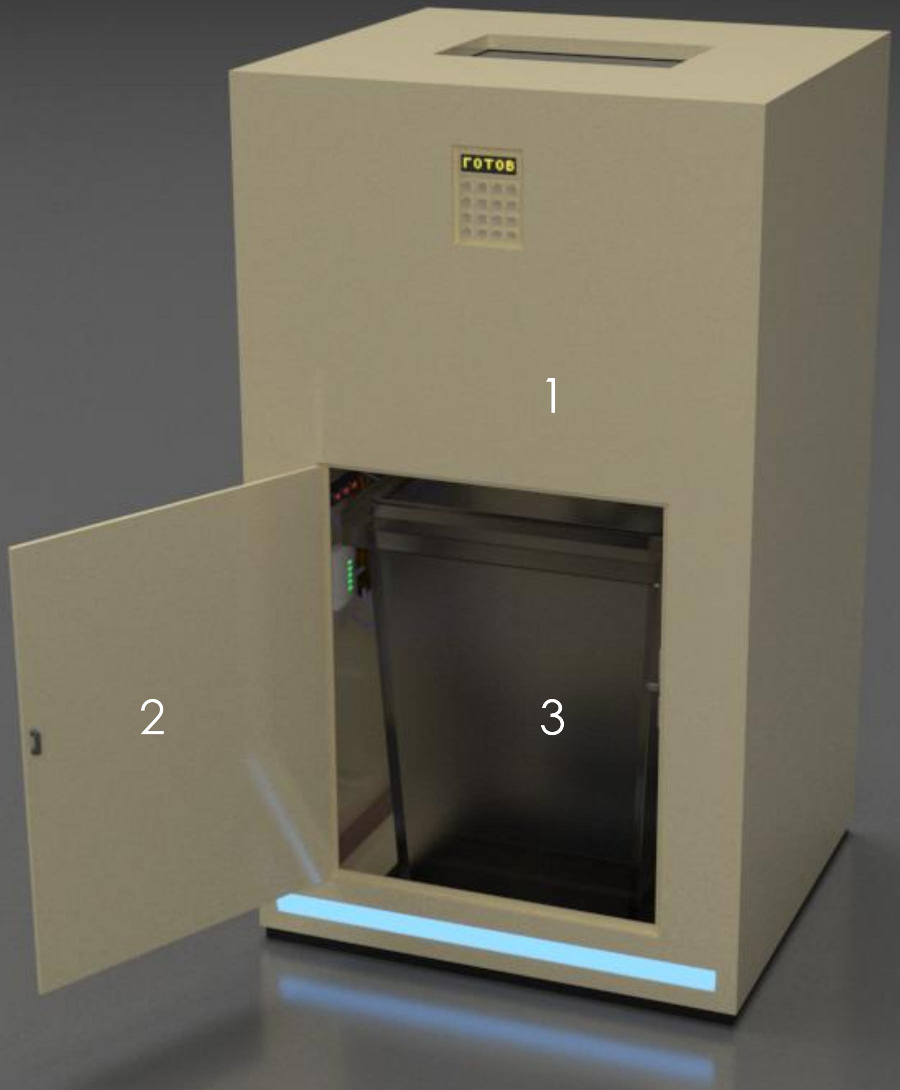
GLASS, BATTERIES

OTHER WASTE

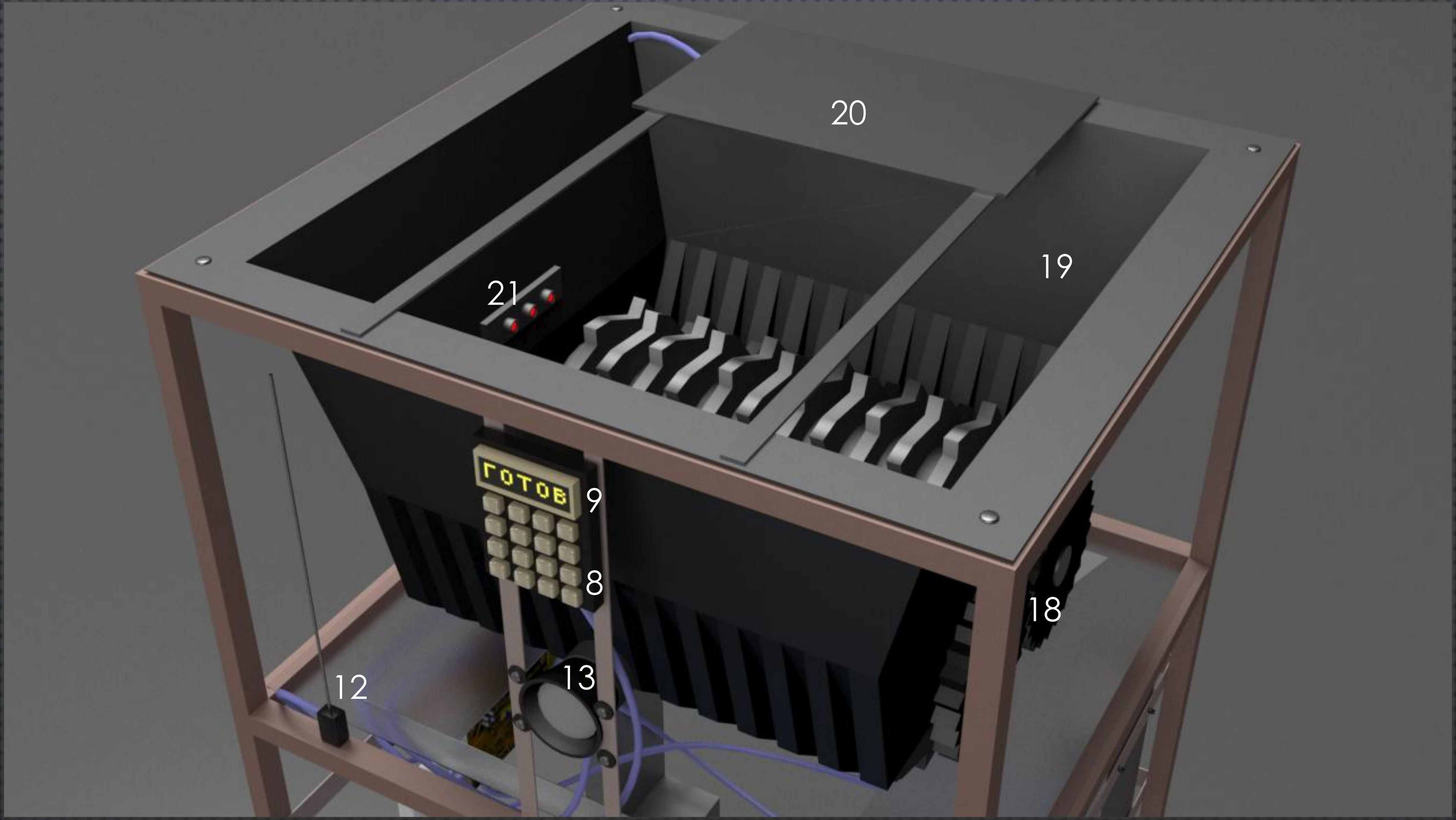
MECHANISM



1. Body
2. Door
3. Container (transparent)
4. Electric door lock
5. Base
6. Neon lamp
7. Metal frame
8. Control buttons
9. Monochrome display
10. Controller board
11. Wireless communication device



12. Antenna
13. Speaker
14. Power supply
15. Fan
16. Electric motor
17. Gearbox
18. Axle with (grinding) drum and gear (2 pcs.)
19. Loading hopper
20. Electric flap
21. Laser for spectral analysis
22. Photosensor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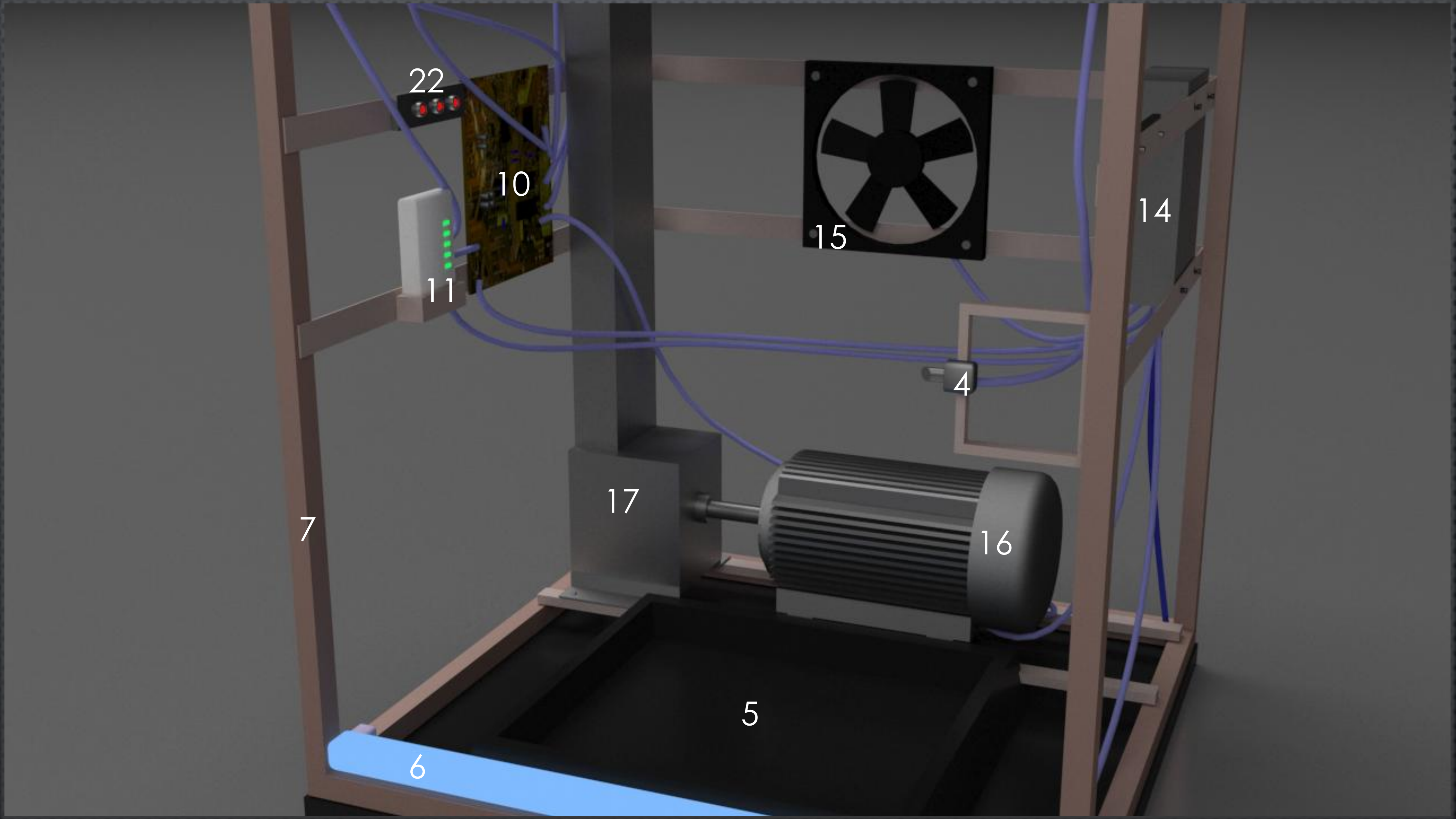
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Operating principle.

1) Metal and plastic recycling.

Let's look at how a plastic recycling machine works.

Approaching the plastic recycling machine, a user enters a code (or a phone number if integrated with third-party services) and confirms. The flap covering the waste entry window opens, and the user dumps the plastic bottles into it. Then, they press the start button. The system uses laser spectral analysis to determine the composition of the waste and, upon confirming that it contains plastic, the flap closes, and the shredding process begins (otherwise, the shredders will not operate). Upon completion, an audible signal is heard and a thank-you message is sent for using the machine.

If the recycling service is integrated into other services (retail chains, housing and utilities), a signal indicating the recycling is complete will be sent to the server of the relevant service via a wireless communication module. Based on the amount of recycled material (calculated via a photo sensor), the user will be credited with bonuses or cash.

When the container is full of recycled material, the photo sensors will detect it as overflowing, and a corresponding message will appear on the display. Also, provided the machine is connected to the network, the operator will see this. When arriving at the site, the worker will enter a command to open the door, empty the overflowing container, and return it to its original location. Metal recycling will follow an identical algorithm (drums for flattening the metal can be installed in place of shredders).

2) Glass and battery recycling.

The glass and battery recycling machine operates similarly to the metal and plastic recycling machine, with one significant exception: it lacks a motor, gearbox, or shredder. Waste is collected in the container in the form in which it was thrown in.

3) Other household waste not classified as garbage.

The machine is a modernized trash bin with an electronic control panel with buttons for opening and closing it. It also contains a plastic waste container, a controller, full sensors, and a power supply.

By implementing this project, we will take another step toward a more humane approach to ecology.

Proper waste disposal remains a distant and incomprehensible concept for many of our citizens. This set of devices will help develop the habit of properly separating waste (perhaps while providing them with some nice little bonuses).

The stylish appearance of these devices will appeal to residents of new buildings, enhancing the appeal of their surrounding areas, and will also fit perfectly into public spaces.

Finally, shredded/compressed waste will allow waste processing plants to obtain more recyclable materials for subsequent recycling.

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